

Project Design Phase-II:

Data Flow Diagram & User Stories

Parameter	Details
Date	08 March 2026
Team ID	
Project Name	VoltEdge: EV Analytics Portal
Maximum Marks	4 Marks

1. Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored.

VoltEdge System Flow:

1. **Data Ingestion:** User selects the raw EV dataset (CSV) to process and load.
2. **PostgreSQL Processing:** The raw data is cleaned and stored in the **public.Tables** schema (e.g., `electric_car_data_clean`).
3. **Tableau Integration:** Cleaned data is extracted from PostgreSQL and passed to Tableau for visualization.
4. **Flask Presentation:** The final interactive dashboards are visualized in the web UI for the end-user.

Note for Word File: You should insert a screenshot of your specific DFD here. Based on the "Simplified Flow" in your template, your DFD should show the flow from **PostgreSQL (Storage)** $\rightarrow$ **Tableau (Processing)** $\rightarrow$ **Flask Web App (Presentation)**.

- This diagram shows the high-level flow of information through your entire system, from data collection to final visualization.

How to use: Paste this image into **Section 1 (Data Flow Diagram)** of your Word file.

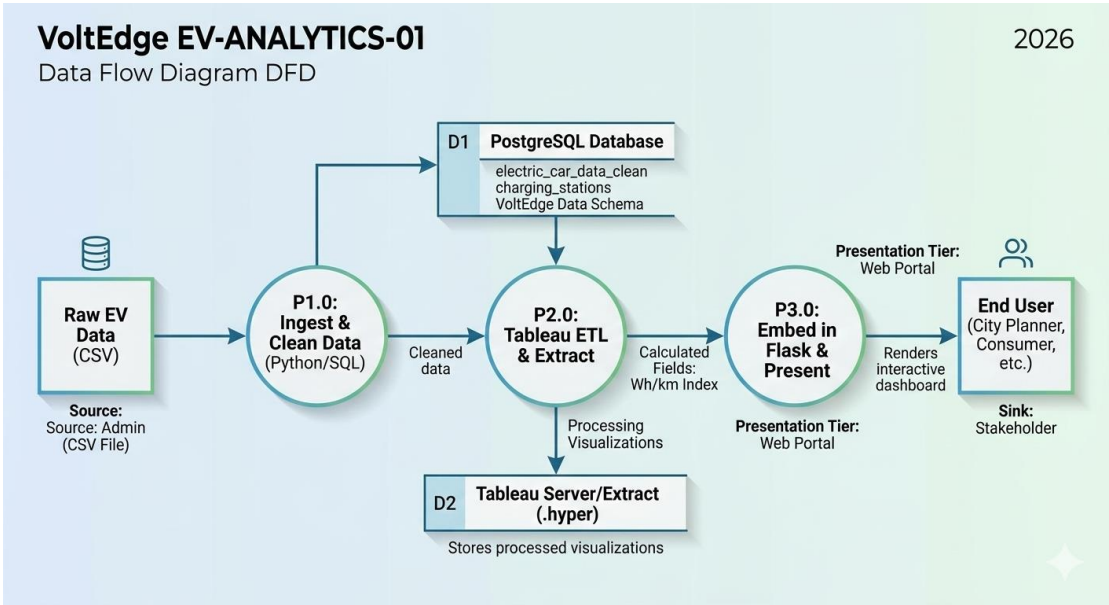


Image Summary: This DFD follows the standard 3-Tier architecture: Data (PostgreSQL) is cleaned (Process 1.0), extracted by Tableau (Process 2.0), and then embedded into the Flask web server (Process 3.0) for the user to view.

2. User Stories

Use the below template to list the user stories for the VoltEdge platform.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Eco-Consumer	Data Exploration	USN-1	As a user, I can filter EVs by brand and body style to find a car that fits my needs.	I can see the list update instantly based on my filters.	High	Sprint-1
Eco-Consumer	Cost Analysis	USN-2	As a user, I can view the "Price-to-Range" ratio to find the most cost-effective EV.	The dashboard displays a "Best Value" ranking.	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
City Planner	Infrastructure Mapping	USN-3	As a planner, I can see a heatmap of existing charging stations to identify gaps.	The map shows clear clusters of station density.	Medium	Sprint-2
Fleet Manager	Efficiency Tracking	USN-4	As a manager, I can compare the Wh/km efficiency of different powertrain types.	I can see a bar chart comparing AWD vs. RWD efficiency.	Medium	Sprint-2
Administrator	Data Management	USN-5	As an admin, I can update the PostgreSQL database with new vehicle models.	New models appear in the Tableau dashboard after a refresh.	Low	Sprint-3

- This image visualizes one of your primary user personas to provide context for the user stories.

How to use: Place this mockup in **Section 2 (User Stories)**, perhaps before the main table, as an "Illustrative User Journey" for the "Eco-Consumer" or "City Planner."

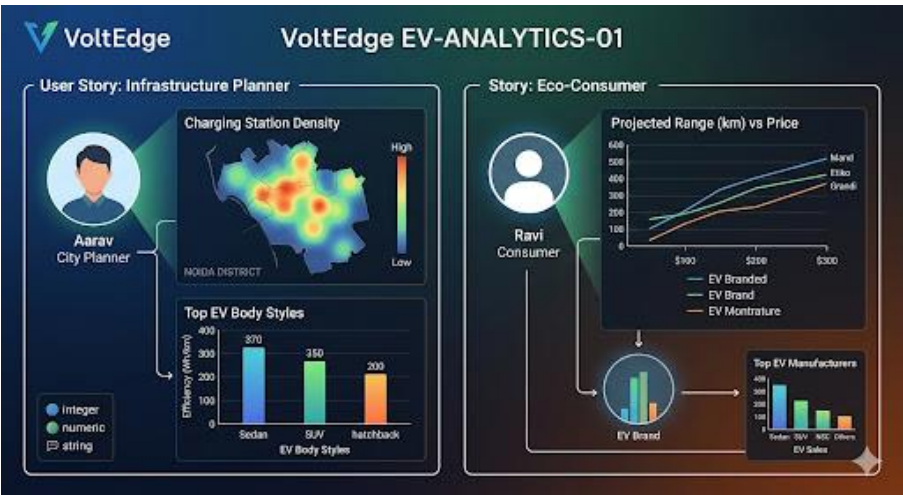


Image Summary: This mockup visualizes the user journey for your personas, with specific charts like the 'Charging Station Density' (Medium Priority User Story USN-3) and 'Price-to-Range' (High Priority User Story USN-2).